

Internet of Things

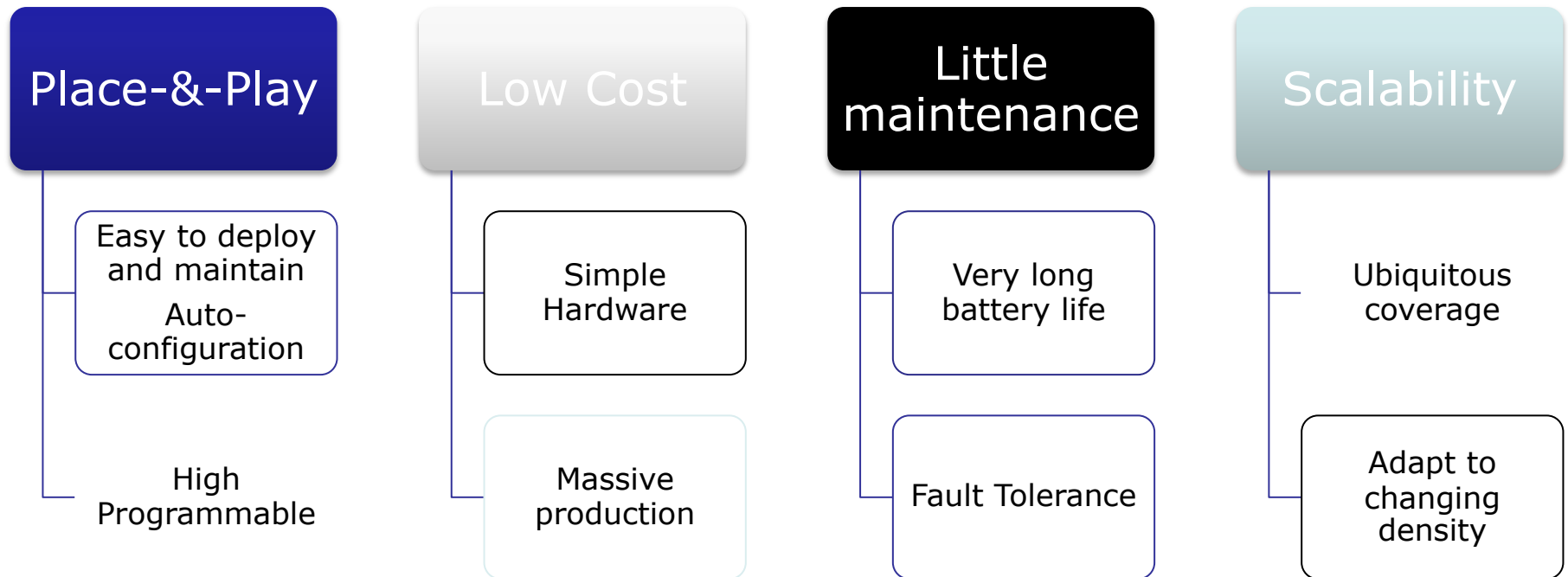
Computer Science Master: IoT Curriculum

IoT Applications and Challenges

Types of IoT applications

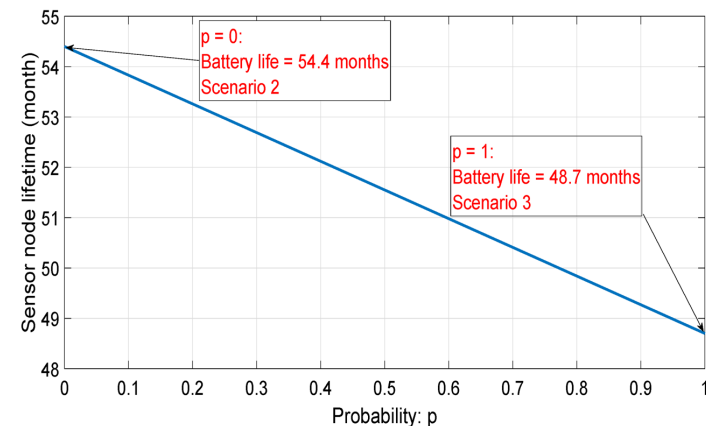
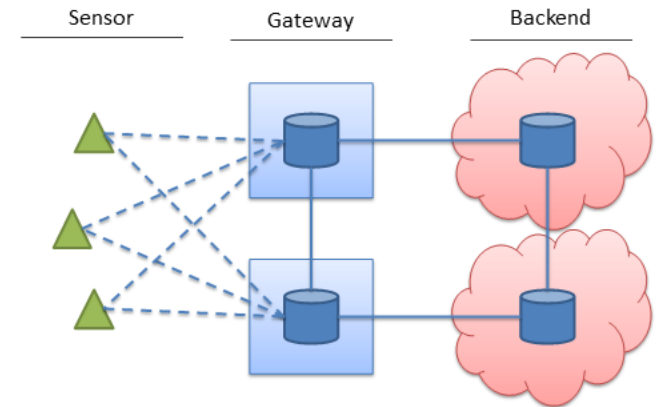
- Event detection
 - Reporting occurrences of events
 - Reporting abnormalities and changes
 - Could require collaboration of other nearby or remote nodes
 - Event definition and classification is an issue
- Periodic measurements
 - Sensors periodically measure and report the observation and measurement data
 - Reporting period is application dependent
- Approximation and pattern detection
 - Sending messages along the boundaries of patterns in both space/time
- Tracking
 - When the source of an event is mobile
 - Sending event updates with location information

IoT Service Requirements



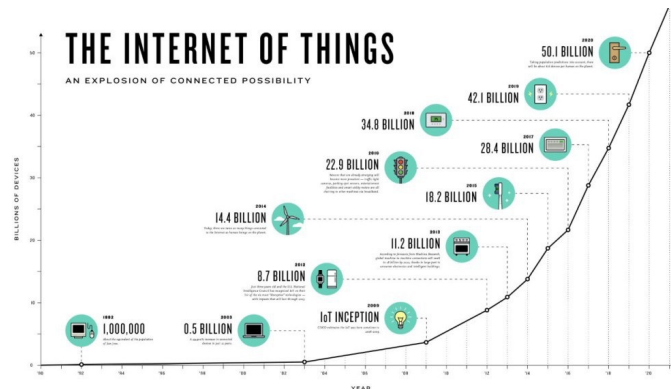
Requirements and challenges

- Fault tolerance
 - The nodes can get damaged, run out of power, the wireless communication between two nodes can be interrupted, etc.
 - To tolerate node failures, redundant deployments can be necessary.
- Lifetime
 - The nodes could have a limited energy supply;
 - Sometimes replacing the energy sources is not practical (e.g. underwater deployment, large/remote field deployments).
 - Energy efficient operation can be a necessity.



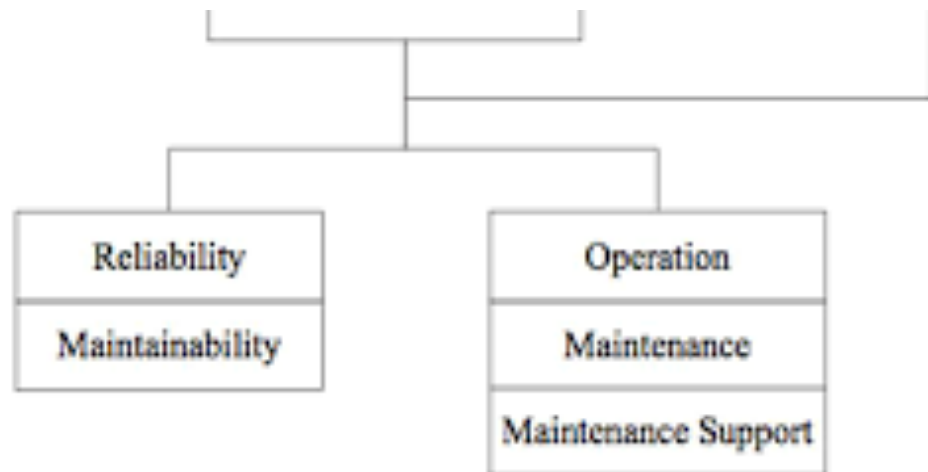
Requirements and challenges – Cont'd

- Scalability
 - An IoT cyber-physical system can consists of a large number of nodes
 - The employed architectures and protocols should scale to these numbers.
- Wide range of densities
 - Density of the network can vary
 - Different applications can have different node densities
 - Density does not need to be homogeneous in the entire network and network should adapt to such variations.



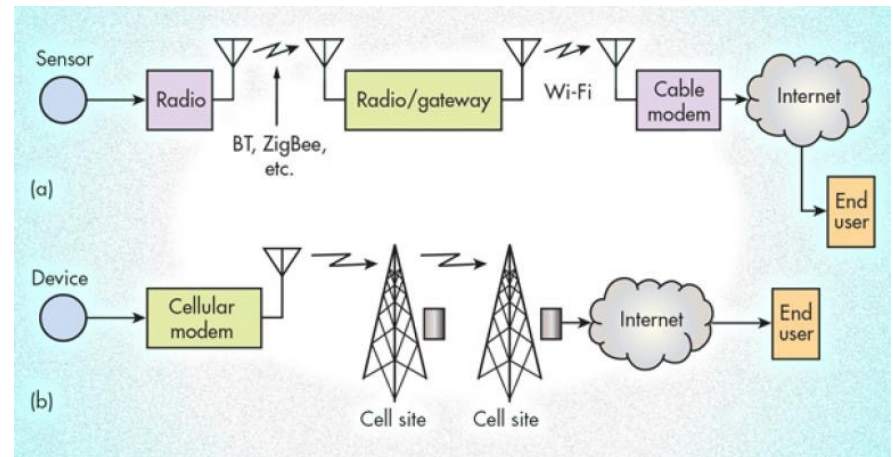
Requirements and challenges – Cont'd

- Programmability
 - Nodes should be flexible and their tasks could change
 - The programmes should be also changeable during operation.
- Maintainability
 - IoT environment can change;
 - The system should be adaptable to the changes.
 - The operational parameters can change to choose different trade-offs (e.g. to provide lower quality when energy efficiency is more important)



Required mechanisms

- Multi-hop wireless communications
 - Communication over long distances can require intermediary nodes as relay (instead of using high transmission power for long range communications).
- Energy-efficient operation
 - To support long lifetime
 - Energy efficient communication/dissemination of information
 - Energy efficient determination of a requested information
- Auto-configuration
 - Self-xxx functionalities
 - Tolerating node failures
 - Integrating new nodes



Required mechanisms

- Collaboration and in-network processing
 - In some applications a single sensor node is not able to handle the given task or provide the requested information.
 - Instead of sending the information from various source to an external network/node, the information can be processed in the network itself.
 - e.g. data aggregation, summarisation and then propagating the processed data with reduced size (hence improving energy efficiency by reducing the amount of data to be transmitted).
- Data-centric
 - Conventional networks often focus on sending data between two specific nodes each equipped with an address.
 - Here what is important is data and the observations and measurements not the node that provides it.

Sensor Data

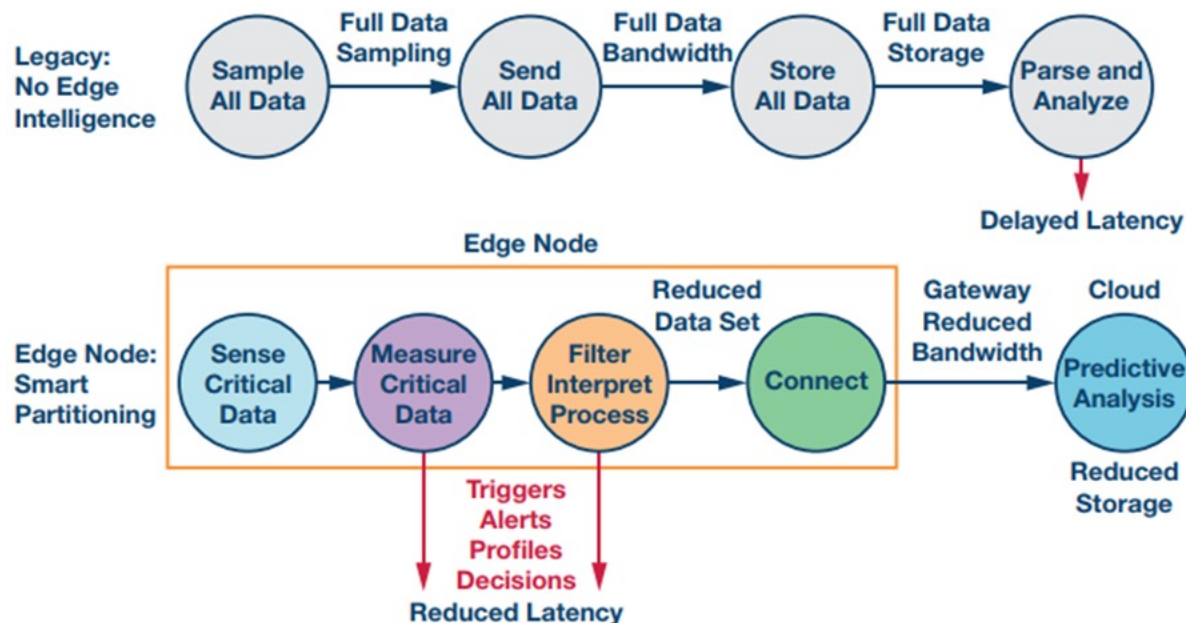
- The sensory data represents physical world observations and measurements and requires time and location and other descriptive attributes to make the data more meaningful.
- For example, a temperature value of 15 degree will be more meaningful when it is described with spatial (e.g. Guildford city centre) and temporal (e.g. 8:15AM GMT, 14-11-2014), and unit (e.g. Celsius) attributes.
- The sensory data can also include other detailed meta-data that describe quality or device related attributes (e.g. Precision, Accuracy).

The benefits of data processing in IoT

- Turn terabytes of Tweets created each day into sentiment analysis related to different events/occurrences or relate them to products and services.
- Convert (billions of) smart meter readings to better predict and balance power consumption.
- Analyze thousands of traffic, pollution, weather, congestion, public transport and event sensory data to provide better traffic and smart city management.
- Monitor patients, elderly care and much more...
- Requires: real-time, reliable, efficient (for low power and resource limited nodes), and scalable solutions.

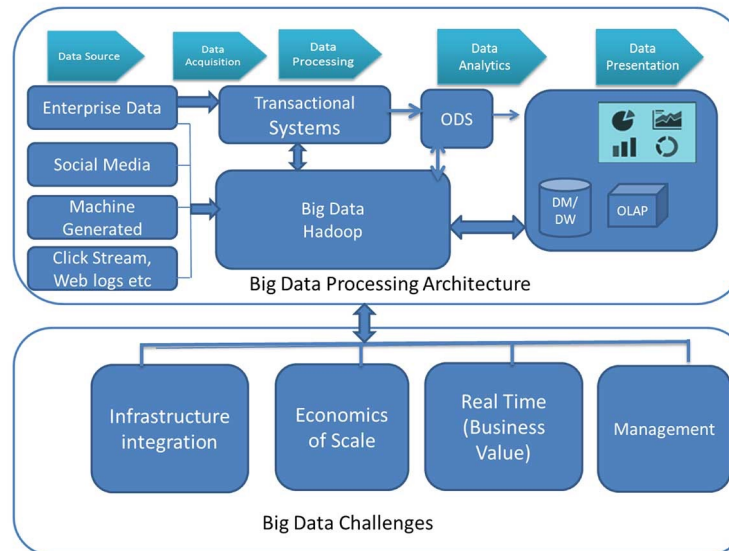
Intelligent Data Processing

- Sensing and data collection, sensor data and data-centric networks
- Access, subscription and integration
- Data processing and stream data analysis
- Query and discovery
- Data classification and clustering



IoT Data Challenges

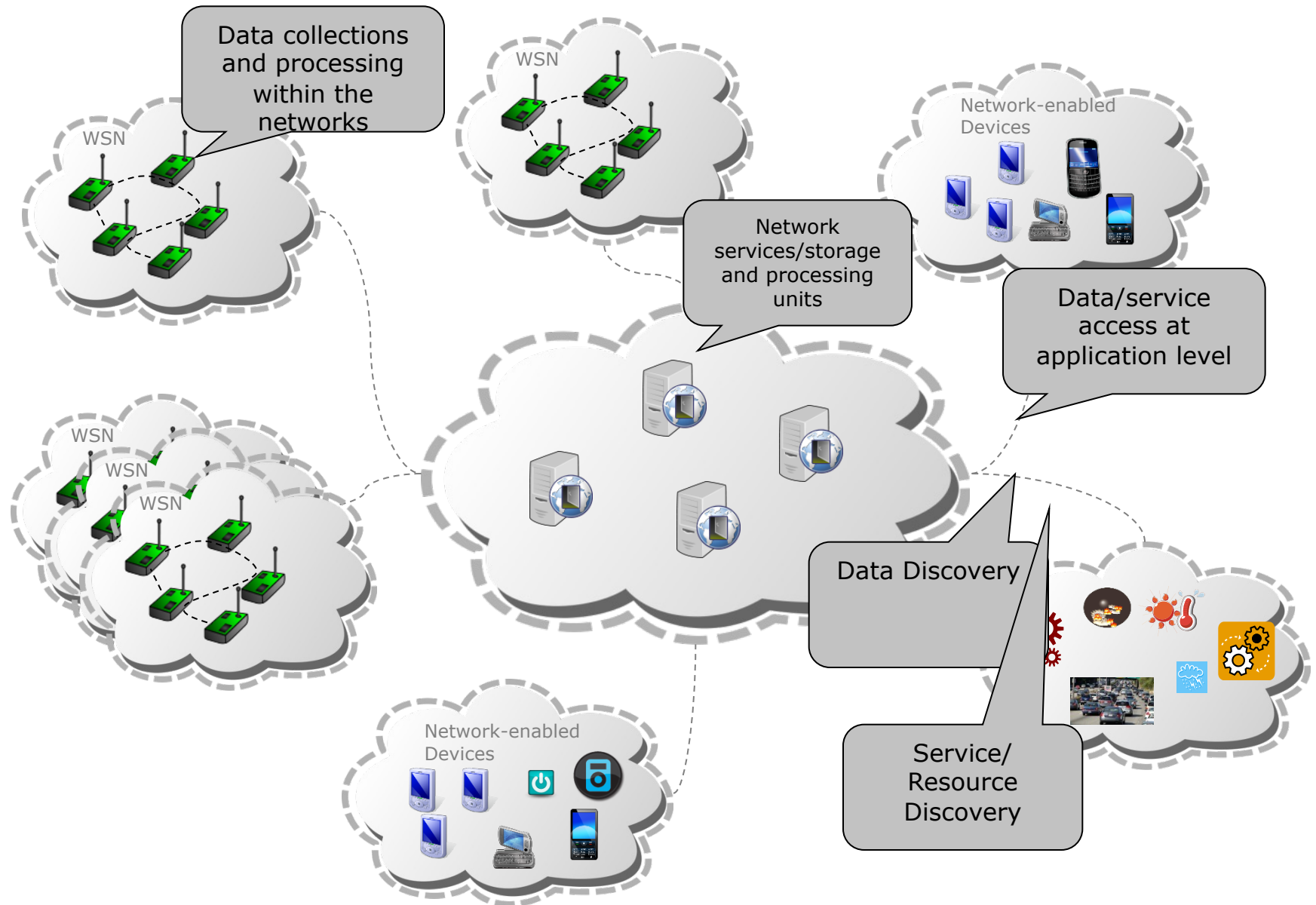
- Multi-modal and heterogeneous
- Noisy and incomplete
- Time and location dependent
- Dynamic and varies in quality
- Crowded sourced data can be unreliable
- Requires (near-) real-time analysis
- Privacy and security are important issues
- Data can be biased- **we need to know our data!**



IoT Data Challenges

- Interoperability: various data in different formats, from different sources (and different qualities)
- Discovery: finding appropriate device and data sources
- Access: Availability and (open) access to resources and data
- Search: querying for data
- Integration: dealing with heterogeneous device, networks and data
- Interpretation: translating data to knowledge usable by people and applications
- Scalability: dealing with large number of devices and myriad of data and computational complexity of interpreting the data.

IoT Data Processing



In-network processing

- Mobile Ad-hoc Networks can be seen as a set of nodes that deliver bits from one end to the other;
- IoT infrastructures on the other end, are expected to provide information, not necessarily original bits
 - Gives additional options
 - e.g., manipulate or process the data in the network
- Main example: aggregation
 - Applying aggregation functions to a obtain an average value of measurement data
 - Typical functions: minimum, maximum, average, sum, ...
 - Not amenable functions: median

In-network processing

- Depending on application, more sophisticated processing of data can take place within the network
 - Example edge detection: locally exchange raw data with neighboring nodes, compute edges, only communicate edge description to far away data sinks
 - Example tracking/angle detection of signal source: Conceive of sensor nodes as a distributed microphone array, use it to compute the angle of a single source, only communicate this angle, not all the raw data
- Exploit **temporal** and **spatial correlation**
 - Observed signals might vary only slowly in time; so no need to transmit all data at full rate all the time
 - Signals of neighboring nodes are often quite similar; only try to transmit differences (details a bit complicated, see later)

Data-centric networking

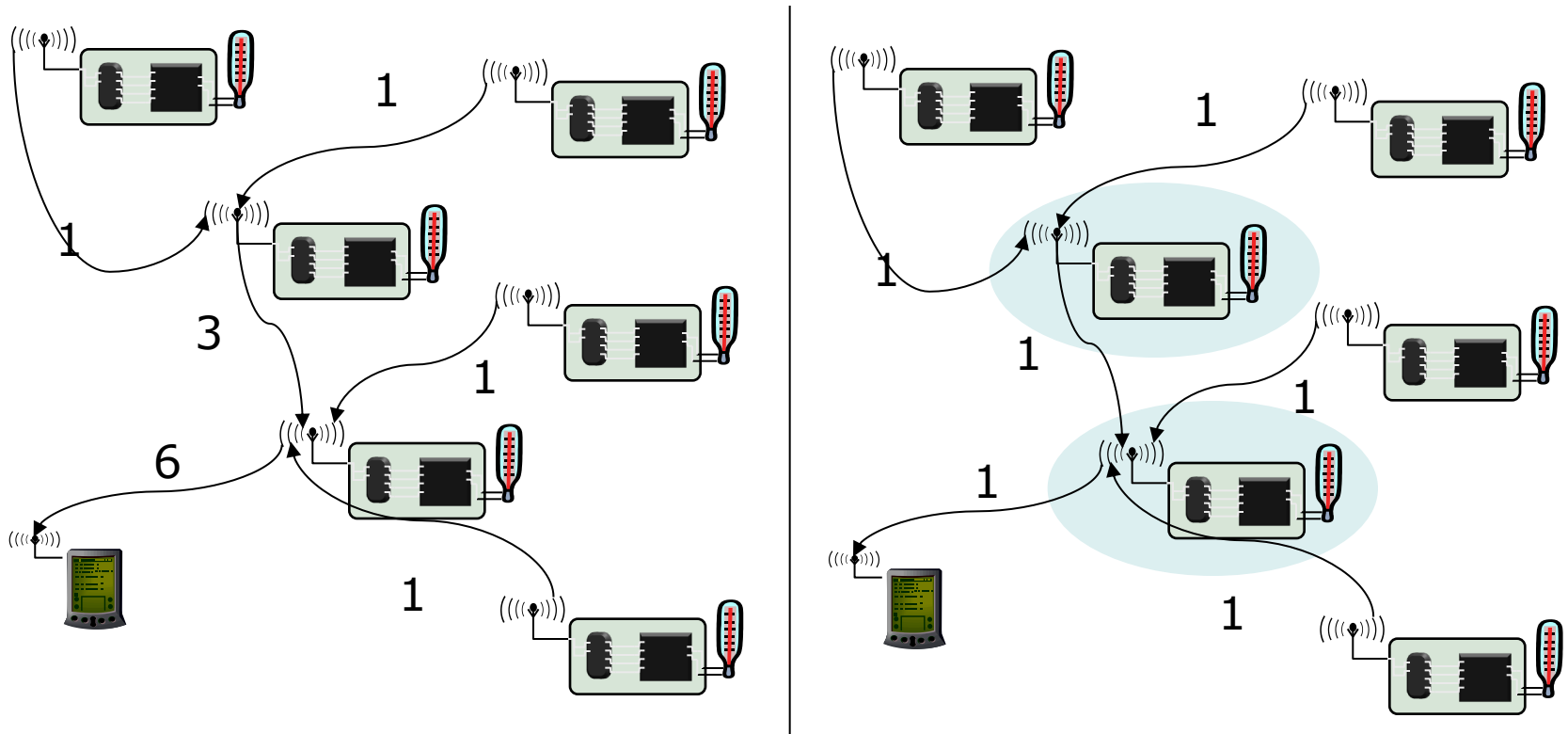
- In typical networks (including ad-hoc networks), network transactions are addressed to the **identities** of specific nodes
 - A “node-centric” or “address-centric” networking paradigm
- In a redundantly deployed sensor networks, specific source of an event, alarm, etc. might not be important
 - Redundancy: e.g., several nodes can observe the same area
- Thus: focus networking transactions on the data directly instead of their senders and transmitters; **data-centric networking**
 - Specially this idea is reinforced by the fact that we might have multiple sources to provide information and observations from the same or similar areas.
 - Principal design change

Data Aggregation

- Computing a smaller representation of a number of data items (or messages) that is extracted from all the individual data items.
- For example computing min/max or mean of sensor data.
- More advance aggregation solutions could use approximation techniques to transform high-dimensionality data to lower-dimensionality abstractions/representations.
- The aggregated data can be smaller in size, represent patterns/abstractions; so in multi-hop networks, nodes can receive data from other node and aggregate them before forwarding them to a sink or gateway.
- Or the aggregation can happen on a sink/gateway node.

Aggregation example

- Reduce number of transmitted bits/packets by applying an aggregation function in the network



Efficacy of an aggregation mechanism

- Accuracy: difference between the resulting value or representation and the original data
 - Some solutions can be lossless or lossy depending on the applied techniques.
- Completeness: the percentage of all the data items that are included in the computation of the aggregated data.
- Latency: delay time to compute and report the aggregated data
 - Computation foot-print; complexity;
- Overhead: the main advantage of the aggregation is reducing the size of the data representation;
 - Aggregation functions can trade-off between accuracy, latency and overhead;
- Aggregation should happen close to the source.